

1st (TS) Philippine Mathematical Olympiad

Area Stage, 31 October 2025 – 7 November 2025

PART I. The answer to each item is an integer from 1 to 999. No solution is needed. Write your answers in the answer sheet. Each correct answer is worth three points.

1. Kaiser's favorite two-digit number is such that subtracting 2 or 4 from it results in a number that shares exactly one digit with the original, but subtracting 3 results in a number that shares no digits with the original. Find Kaiser's favorite number.
2. A set of primes is called *complete* if the digits of its elements are $1, 2, \dots, 9$ in some order. For example, the set $\{7, 83, 421, 659\}$ is complete. Compute the smallest possible sum of elements of a complete set.
3. In parallelogram $ABCD$, the angle bisectors of $\angle A$ and $\angle B$ intersect segment CD and divide the interior of the parallelogram into four regions: two triangles and two quadrilaterals. Given that the triangles have areas 27 and 75, find the area of $ABCD$.
4. Compute the number of 6-digit positive integers $\overline{ABCDEF}$ satisfying:
 - $\overline{AB}$ is divisible by 4.
 - $\overline{ABCD}$ is divisible by 16.
 - $\overline{ABCDEF}$ is divisible by 64.
5. The graphs of $y = x^3 - 5x^2 + 7x + 4$ and $y = 2x^2 - 6$ intersect at A , B , and C . If the sum of the coordinates of the centroid of triangle ABC can be expressed as $\frac{m}{n}$ for relatively prime positive integers m and n , find $m + n$.
6. In regular polygon $A_1A_2 \cdots A_n$ for a positive integer $n \geq 25$, define X as the reflection of A_1 across A_3 . Given that X , A_5 , and A_{25} are collinear, find n .
7. The positive divisors of 1001 are randomly written on four two-sided cards, one on each face. If the probability that on every card, one number divides the other can be expressed as $\frac{m}{n}$, find $m + n$.
8. Let ABC be a triangle with centroid G . Suppose the reflection of A across B lies on the circumcircle of BGC such that $GB = 3$ and $GC = 9$. If the area of ABC can be expressed as $m\sqrt{n}$ for square-free n , find $m + n$.
9. Each tile of a 3×3 chessboard is colored either white or black at random. If the probability that a knight can move from the bottom left tile to the top right tile without changing colors or a knight can move from the bottom right tile to the top left tile without changing colors, but not both can be expressed as $\frac{m}{n}$ for relatively prime positive integers m and n , find $m + n$.
10. How many integers $1 < n \leq 1000$ satisfy the property that $p^2 + 2$ divides n , where p is the smallest prime factor of n ?
11. Let ABC be a triangle with $AB = 13$, $BC = 14$, $CA = 15$, and circumcenter O . Suppose AO intersects the circumcircle of BOC at $X \neq O$. If the length of AX can be expressed as $\frac{m}{n}$ for relatively prime positive integers m and n , find $m + n$.
12. Let n be the smallest positive integer such that

$$3 \left\lfloor \frac{n}{3} \right\rfloor - 4 \left\lfloor \frac{n}{4} \right\rfloor = 7 \left\lfloor \frac{n}{7} \right\rfloor - 8 \left\lfloor \frac{n}{8} \right\rfloor = 11 \left\lfloor \frac{n}{11} \right\rfloor - 12 \left\lfloor \frac{n}{12} \right\rfloor = 1.$$

Find the remainder when n is divided by 1000.

13. Let f be a function defined by $f(1) = 1$, $f(30) = 0$, and for all $x \notin \{1, 30\}$,

$$f(x) = \sum_{\text{prime } p|x} f\left(\frac{x}{p}\right).$$

Find the remainder when $f(1620000)$ is divided by 1000.

14. Acute triangle ABC has circumcenter O and circumradius 81. Let line ℓ be the tangent to the circumcircle of ABC at A such that the distances from B and C to ℓ are 64 and 144 respectively. If the area of ABC can be expressed as $m\sqrt{n}$ for square-free n , find the remainder when $m + n$ is divided by 1000.
15. For each nonnegative integer k , $P_k(x)$ is a real polynomial of degree k with leading coefficient $\frac{k+1}{k!}$. For each k , we have that

$$P_k(k) = P_{k+1}(k) = P_{k+2}(k) = \cdots$$

Given that $P_{100}(100)$ is an integer, compute the number of positive integer divisors of $P_{100}(100)$.

PART II. Write your solutions to each problem in the solution sheets. Each complete and correct solution is worth ten points.

- Chick and Rick play a game on n points spaced equally on a circle. Chick goes first, and on each turn the player draws a chord on the circle between some pair of the points. The first player to draw a chord intersecting an existing chord strictly in the interior of the circle loses. Find, with proof, all positive integers n such that Chick wins when the game is played optimally.
- Find, with proof, a non-recursive formula for the number of sequences $s_1, s_2, \dots, s_N$ such that $s_1 = 1$, $s_2 = 2$, $s_3 = 3$, and for all integers $3 < k \leq N$,

$$s_k \in \{s_{k-1}, s_{k-2}, s_{k-3}\}.$$

- A sequence $\{a_n\}_{n \geq 1}$ of positive integers is defined recursively by $a_1 = k$ and

$$\begin{cases} a_{n+1} = \frac{1}{2} \cdot a_n & \text{if } a_n \equiv 0 \pmod{2} \\ a_{n+1} = a_n + 77 & \text{if } a_n \equiv 1 \pmod{2} \end{cases}$$

for all n . Determine, with proof, the sum of all values of k for which $a_6 = a_1$.